Skill and practice signatures across tasks

16 contrasts in 10 datasets prepared by DataPrep; 1,350,049 trials. Change = B relative to A.

Prediction tested (SKILL_vs_PRACTICE_explained.pdf, p. 5)	Measured	Answer
1. Time on task is front-led (10th percentile falls more than the 99th), as practice predicts	1 of 8 contrasts	a few
2. Time on task: mu falls and tau does not fall, as practice predicts	3 of 8 contrasts	a few
3. Word frequency (more stored instances) is front-led	F1: tail-led	tail-led
4. Expertise is tail-led with mu rising, as skill (drive precision) predicts	0 of 2 contrasts	none
5. Stimulus strength moves mu and tau in the same direction (the document's drive-level axis)	1 of 2 contrasts	one

1. Front minus tail = percent change at the 10th percentile minus that at the 99th; front-led: 95% interval below 0; tail-led: above 0; uniform: includes 0. The document's simulation gives -42 for practice and +50 for skill.

2. Direction of mu, sigma, tau: 95% interval of the median percent change below 0 (falls), above 0 (rises), or including 0. Intervals: bootstrap over units and trials (percentiles) or over units (ex-Gaussian), 1,000 replicates, seed 61106.

3. Trials: DataPrep's usable rows, correct trials where accuracy is recorded, test phase only. Answer categories: none = 0; most = more than half; a few = between.

4. Sources: 4_output/contrast_summary.csv, unit_changes.csv, unit_measures.csv.

Step 1. How each dataset tests the hypothesis

Each contrast moves one of the document's axes and holds the task fixed, so each is an independent test.

Id	Dataset (task)	Axis	A vs B	Units	Prediction
P1	ratcliff1998 (brightness discrimination)	Time on task	first quarter of sessions vs last quarter	3	front-led; mu, sigma fall; tau flat
P2	wagenmakers2008 (lexical decision)	Time on task	first quarter of trials vs last quarter	17	front-led; mu, sigma fall; tau flat
P3	keuleers2012 (lexical decision (BLP))	Time on task	first quarter of trials vs last quarter	78	front-led; mu, sigma fall; tau flat
P4	cochrane2023 (motion discrimination)	Time on task	day 1 (first quarter) vs day 4 (last quarter)	21	front-led; mu, sigma fall; tau flat
P5	wolfe2010 (visual search)	Time on task	first quarter of trials vs last quarter	29	front-led; mu, sigma fall; tau flat
P6	rouder2008 (visual working memory)	Time on task	first quarter of trials vs last quarter	23	front-led; mu, sigma fall; tau flat
P7	pratte2010 (Stroop and Simon)	Time on task	first quarter of trials vs last quarter	153	front-led; mu, sigma fall; tau flat
P8	enkavi2019_threebytwo (cued task switching)	Time on task	first quarter of trials vs last quarter	521	front-led; mu, sigma fall; tau flat
P9	enkavi2019_ant (attention network (flanker))	Time on task	first quarter of trials vs last quarter	522	front-led; mu, sigma fall; tau flat
A1	ratcliff1998 (brightness discrimination)	Speed vs accuracy	speed instruction vs accuracy instruction	3	benchmark: no prediction
A2	wagenmakers2008 (lexical decision)	Speed vs accuracy	speed instruction vs accuracy instruction	17	benchmark: no prediction
F1	wagenmakers2008 (lexical decision)	Word frequency	very low frequency words vs high frequency words	17	front-led; mu, sigma fall; tau flat
S1	sc2_action_type3 (StarCraft 2 by league)	Expertise	Bronze games vs Grandmaster games	167 / 32	tail-led; mu rises; tau falls
S2	cochrane2023 (motion discrimination)	Expertise	non-players, day 1 vs action video game players, day 1	10 / 11	tail-led; mu rises; tau falls
D1	roitman2002 (motion discrimination (monkeys))	Stimulus strength	coherence 0.032 vs coherence 0.512	2	mu and tau move together
D2	cochrane2023 (motion discrimination)	Stimulus strength	coherence 0.064 vs coherence 0.448	21	mu and tau move together

1. Time on task (P1-P8): each unit's trials in run order, first vs last quarter, within cells that alternate by block (speed/accuracy instruction, search type, Stroop/Simon). Other conditions are randomised within blocks; their share differs by at most 1.0 point between quarters (process/output/check_02.txt).

2. Word frequency (F1): lifetime exposure; instance theory treats frequent words as more stored instances (Logan, 1988).

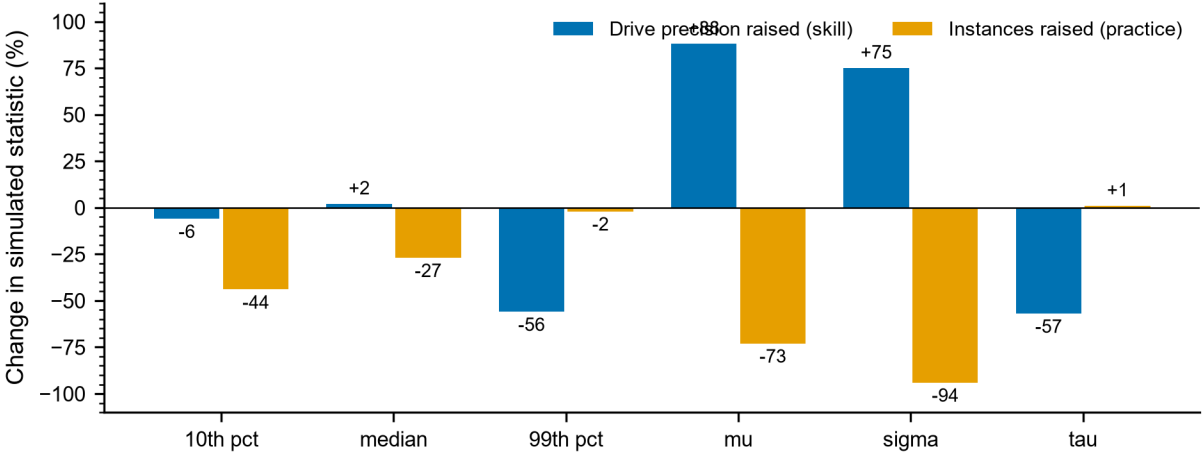


Figure 1. Predictions as reported in SKILL_vs_PRACTICE_explained.pdf (p. 4): percent change in each statistic of the extended Wong-Wang model's RT distribution when drive precision (skill) or the number of stored instances (practice) is raised, 20,000 simulated trials each. Not recomputed here.

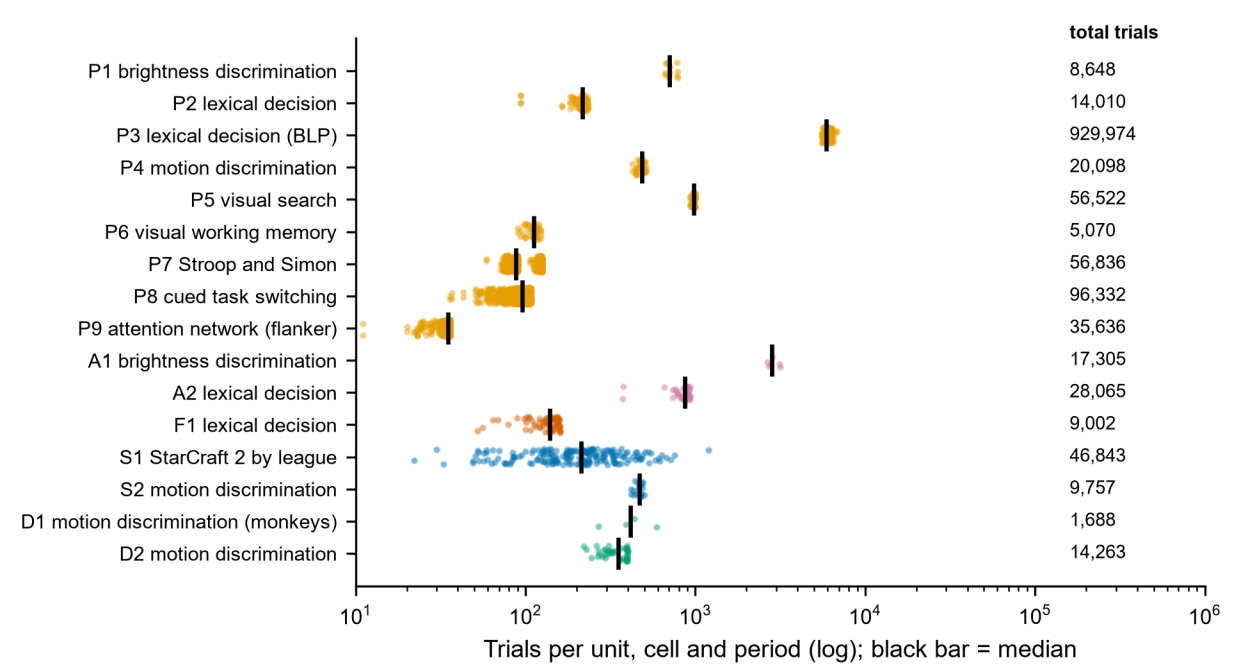


Figure 2. Step 2, the data. Trials behind each percentile estimate: one point per unit, cell and period. Below about 100 trials, the 99th percentile rests on one or two trials (P6, P8, and many StarCraft 2 games in S1).

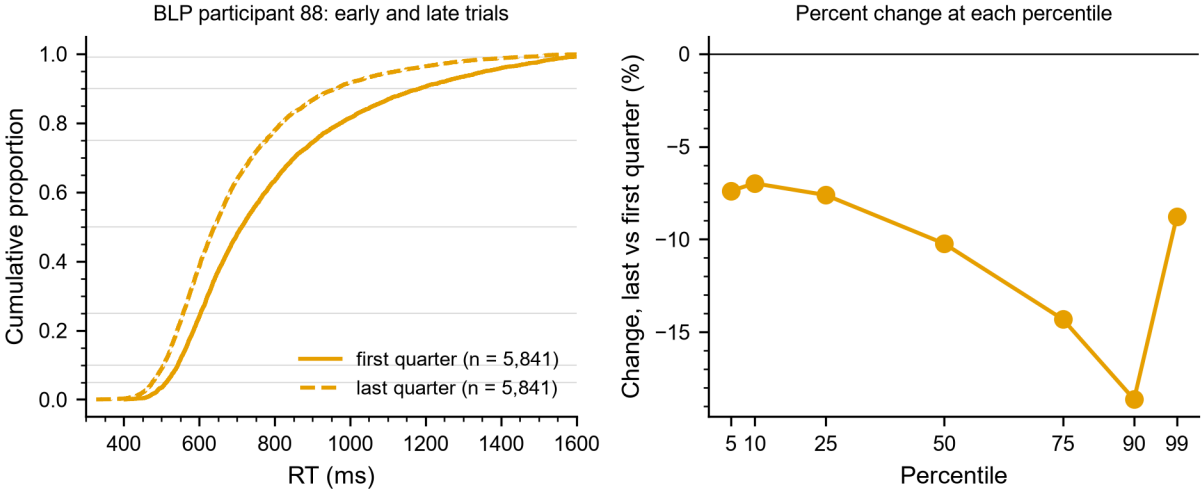


Figure 3. Step 3, the method, on one unit: the participant with the median change in median RT in the British Lexicon Project (P3). Left: distribution of correct RTs in the first and last quarter of trials. Right: percent change at each percentile. Practice predicts the largest fall at low percentiles and none at the 99th.

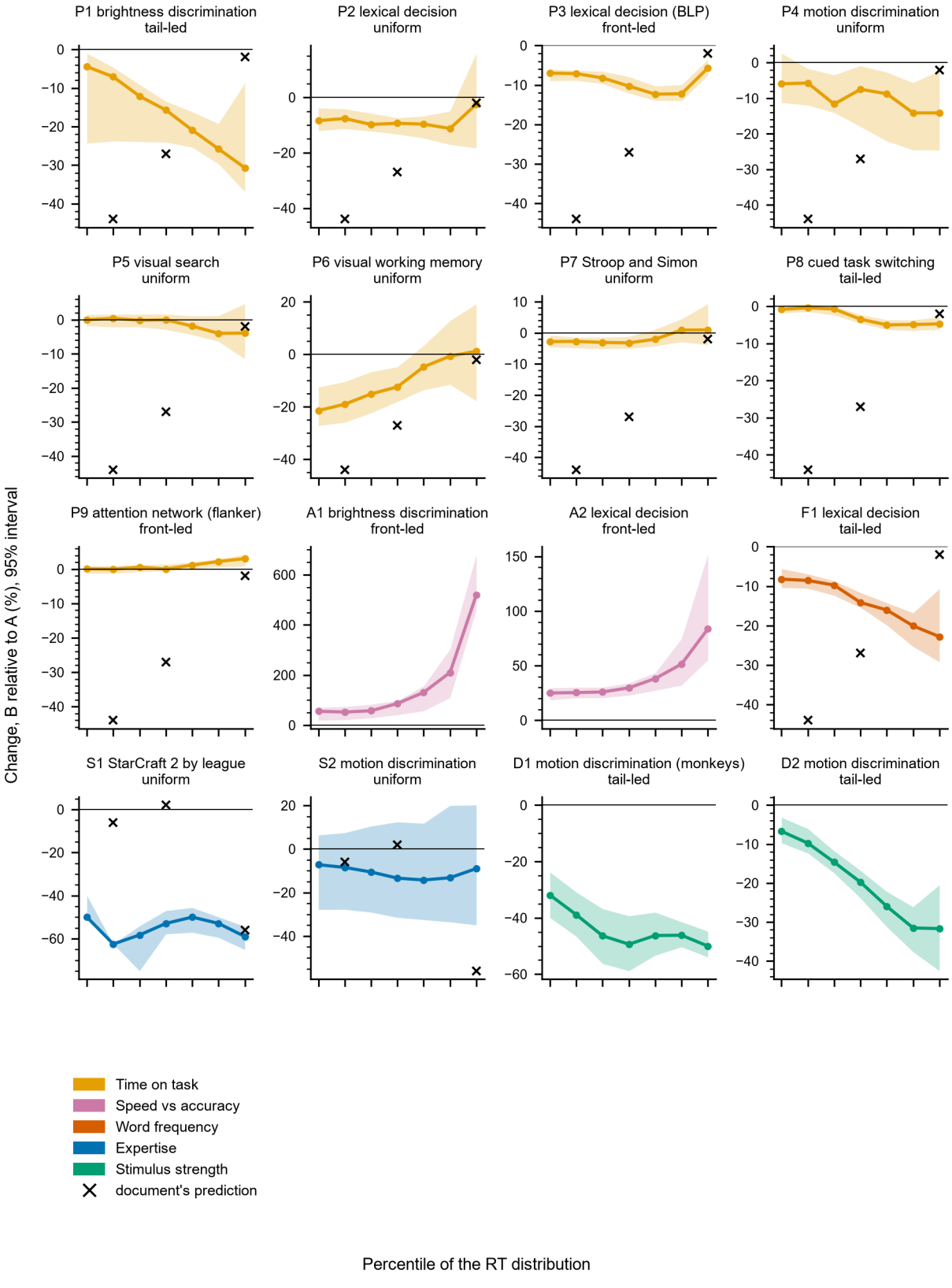


Figure 4. Step 4, results. Percent change at each percentile for every contrast, with 95% bootstrap intervals (bands); the label under each title is the classification of front minus tail. Crosses: the document's simulated change at the 10th, 50th and 99th percentiles for the mechanism the contrast moves. Each panel has its own y scale.

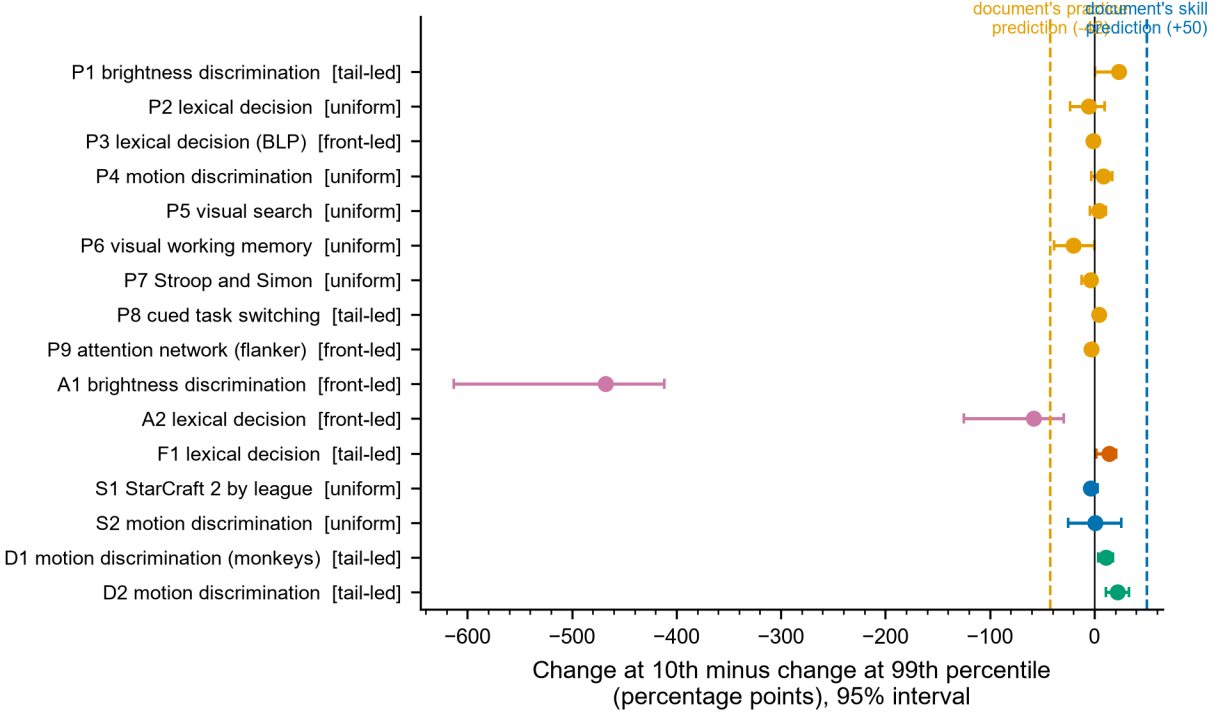


Figure 5. Where the change lives, one row per contrast. Below 0: the fast front changed more than the slow tail (front-led, the practice signature); above 0: the tail changed more (tail-led, the skill signature). Dashed: the document's simulated values. Colour: axis, as in Figure 4.

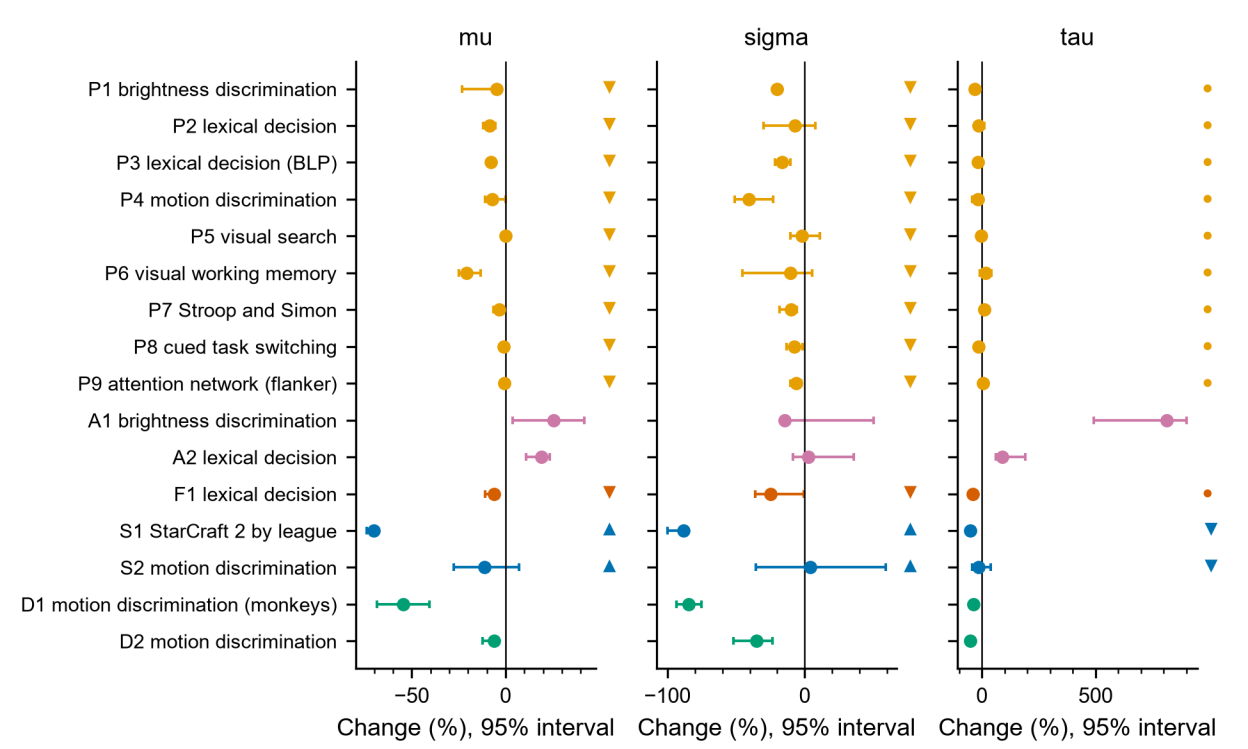


Figure 6. Median percent change in ex-Gaussian mu, sigma and tau, with 95% intervals over units. Symbols right of each panel: the direction the document predicts (▲ rises, ▼ falls, ● no change of more than 5%). With 2 or 3 units (D1, P1) the interval over units is not informative, and where a unit has no defined change (D1 tau, monkey 2) none is drawn.

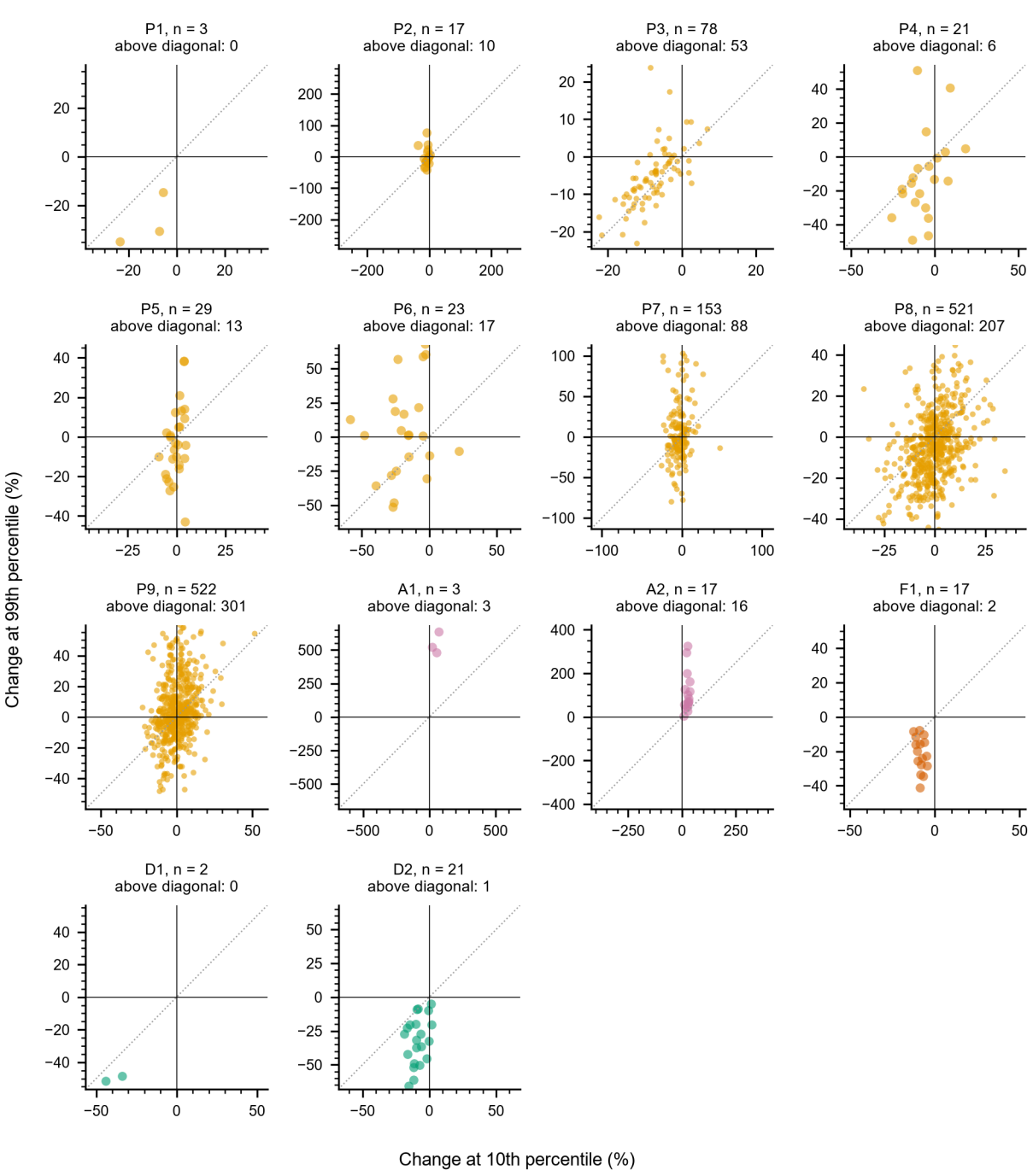


Figure 7. Each unit's change at the 10th against the 99th percentile, within-unit contrasts. Above the dotted diagonal (count in each title), the 10th percentile changed more (more negative) than the 99th. Axes are symmetric and set per panel; the most extreme 2% of values may fall outside.

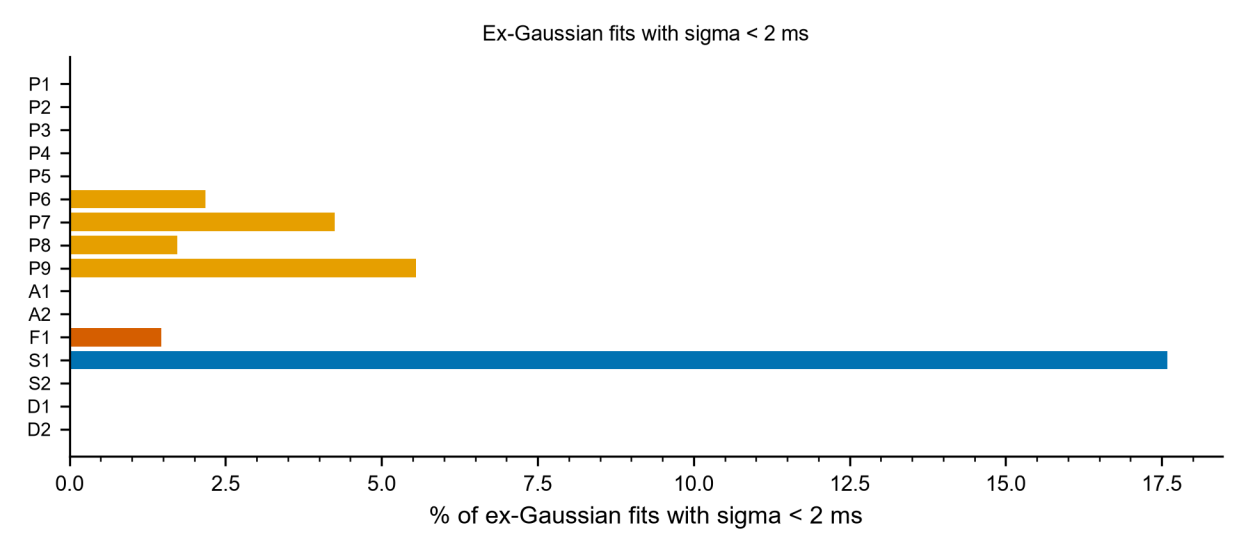


Figure 8. Share of ex-Gaussian fits whose sigma is under 2 ms, where mu sits at the fastest RTs and is not a separate location; these fits are kept. Checks run separately: pooled conditions differ by at most 1.0 point in share between first and last quarter (process/output/check_02.txt); fit recovery on simulated data (check_01.txt).